

## Organisms Transfer Energy and Matter in Ecosystems (Chapter 55)

**8** Primary producers convert the energy in sunlight to chemical energy through photosynthesis. Their growth is often limited by abiotic factors such as low temperatures, scarce soil nutrients, and lack of light in winter. (See Figures 10.6, 52.13, and 55.4.)

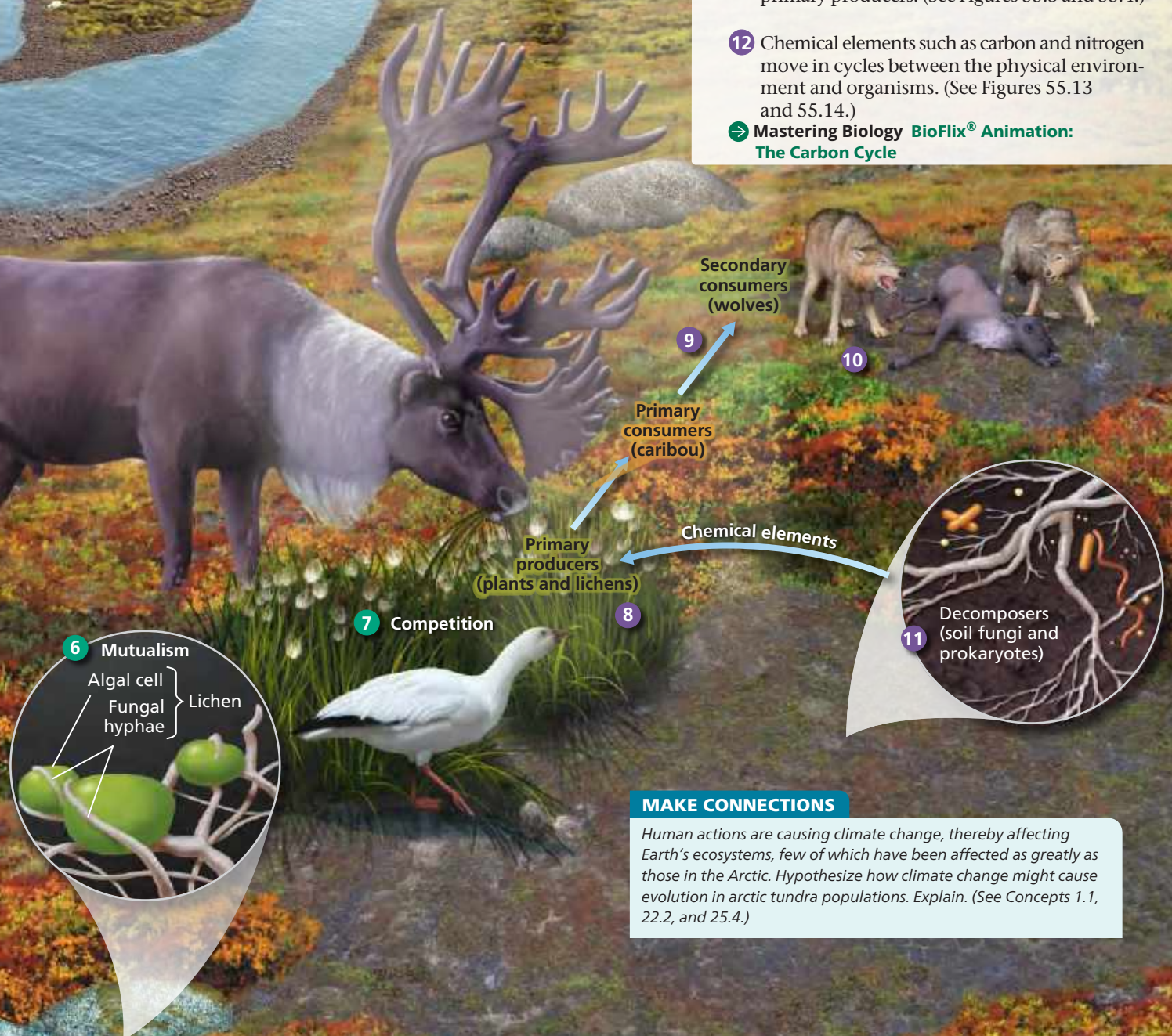
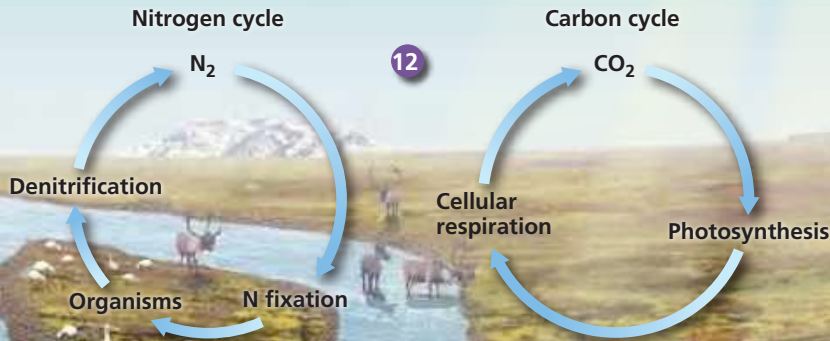
**9** Food chains are typically short in the tundra because primary production is lower than in most other ecosystems. (See Figure 54.16.)

**10** Energy flows through ecosystems. When one organism eats another, only 10% of the energy transfers from one trophic level to the next. (See Figure 55.10.)

**11** Decomposers recycle chemical elements back to primary producers. (See Figures 55.3 and 55.4.)

**12** Chemical elements such as carbon and nitrogen move in cycles between the physical environment and organisms. (See Figures 55.13 and 55.14.)

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### MAKE CONNECTIONS

Human actions are causing climate change, thereby affecting Earth's ecosystems, few of which have been affected as greatly as those in the Arctic. Hypothesize how climate change might cause evolution in arctic tundra populations. Explain. (See Concepts 1.1, 22.2, and 25.4.)